

Preserving Molecular Information During Pathology Report De-identification: A Synthetic Benchmark

Background: De-identified pathology reports could support model-assisted standardization, summarization, hypothesis generation and testing, and research abstraction. Preparing reports for these uses requires removing identifiers while preserving molecular findings.

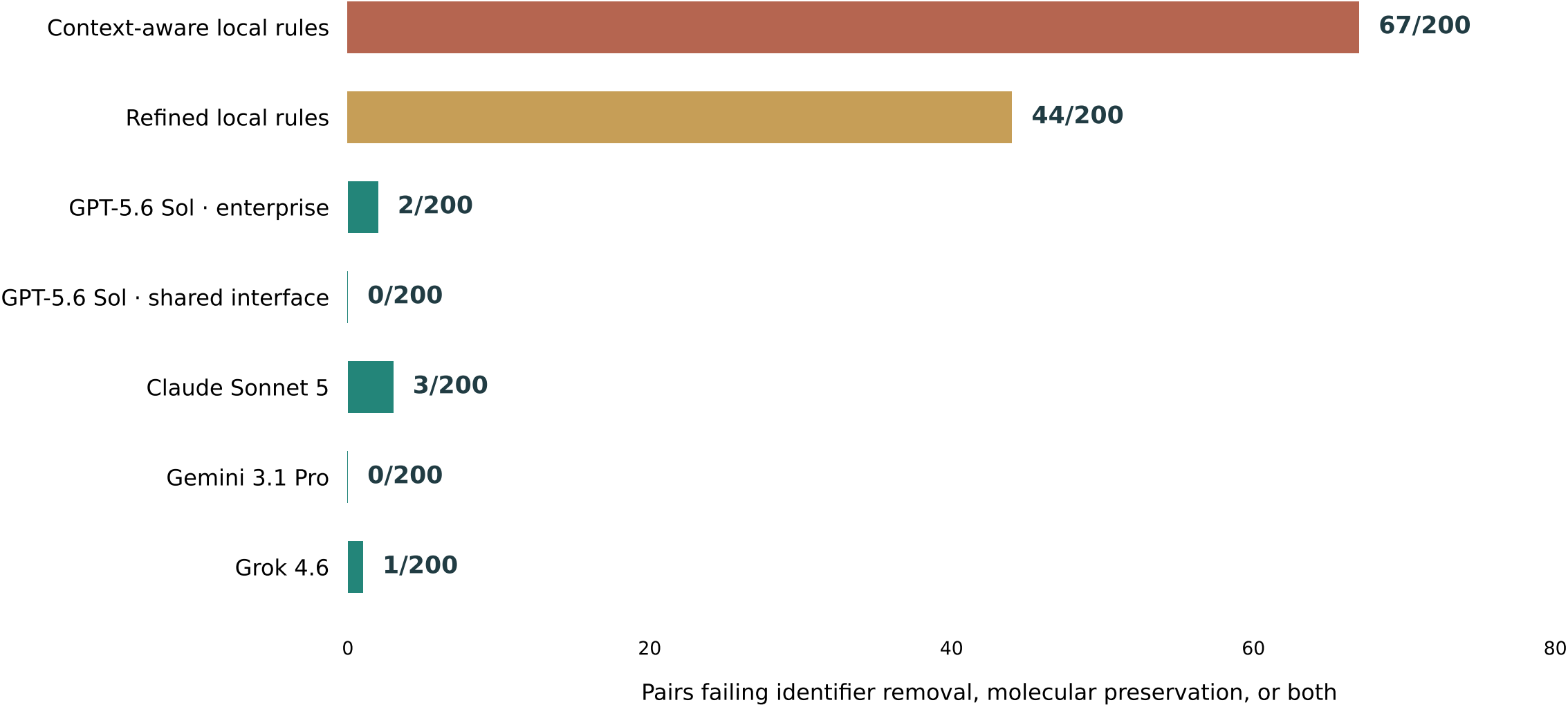
Design: We compared an institution-developed pipeline, context-aware rules, and refined context-aware rules with five frontier LLM configurations (GPT-5.6 Sol through Enterprise and a shared model-access interface; Claude Sonnet 5, Gemini 3.1 Pro, and Grok 4.6 through that same interface). Synthetic report pairs tested molecular-expression preservation and removal of similarly formatted identifying digits (200 pairs). A separate experiment used investigator-designed report layouts targeting known weaknesses and contextual challenges (eight patterns; 25 pairs each). Exploratory analyses compared paired outcomes and grouped reports sharing patterns.

Results: LLM configurations achieved both objectives more consistently than the institution-developed pipeline or either rule method (197-200/200 versus 0/200, 133/200, and 156/200, respectively). Molecular preservation was complete with LLMs, incomplete with both rule methods, and absent with the pipeline (200/200, 159/200, and 0/200). Residual identifier exposure was uncommon with refined rules and LLMs (3/200 and 0-3/200). Pair-level comparisons favored LLMs, but pattern-grouped analyses were inconclusive (adjusted $p < .001$; $p = .156$ versus earlier rules and $.375$ versus refined rules). Local failures included molecular over-redaction and identifiers missed across line breaks, separated footers, parentheses, and modified labels. In the separate experiment, refinement halved identifier exposures, with parenthetical and modified-label failures persisting (100/200 to 50/200).

Conclusions: Local refinement improved identifier removal without resolving molecular information loss. LLM configurations performed better descriptively; safety and performance on independent clinical reports remain unestablished.

1. Fewer failed pairs with LLM configurations

200 unique synthetic pairs per approach | Failure of identifier removal, molecular preservation, or both



PAIR-LEVEL COMPARISON

Adjusted p < .001 versus either local rule method

ACCOUNTING FOR SHARED REPORT PATTERNS

Adjusted p = .156 / .375; evidence is inconclusive

Exploratory comparisons. Shared patterns limit the independence of the 200 pairs. Pattern-grouped values compare LLMs with earlier / refined rules. All non-Enterprise models used the same shared interface. The institution-developed pipeline failed all 200 pairs (Supplementary Figure S1).

Lung adenocarcinoma: preserve the finding, remove the identifiers

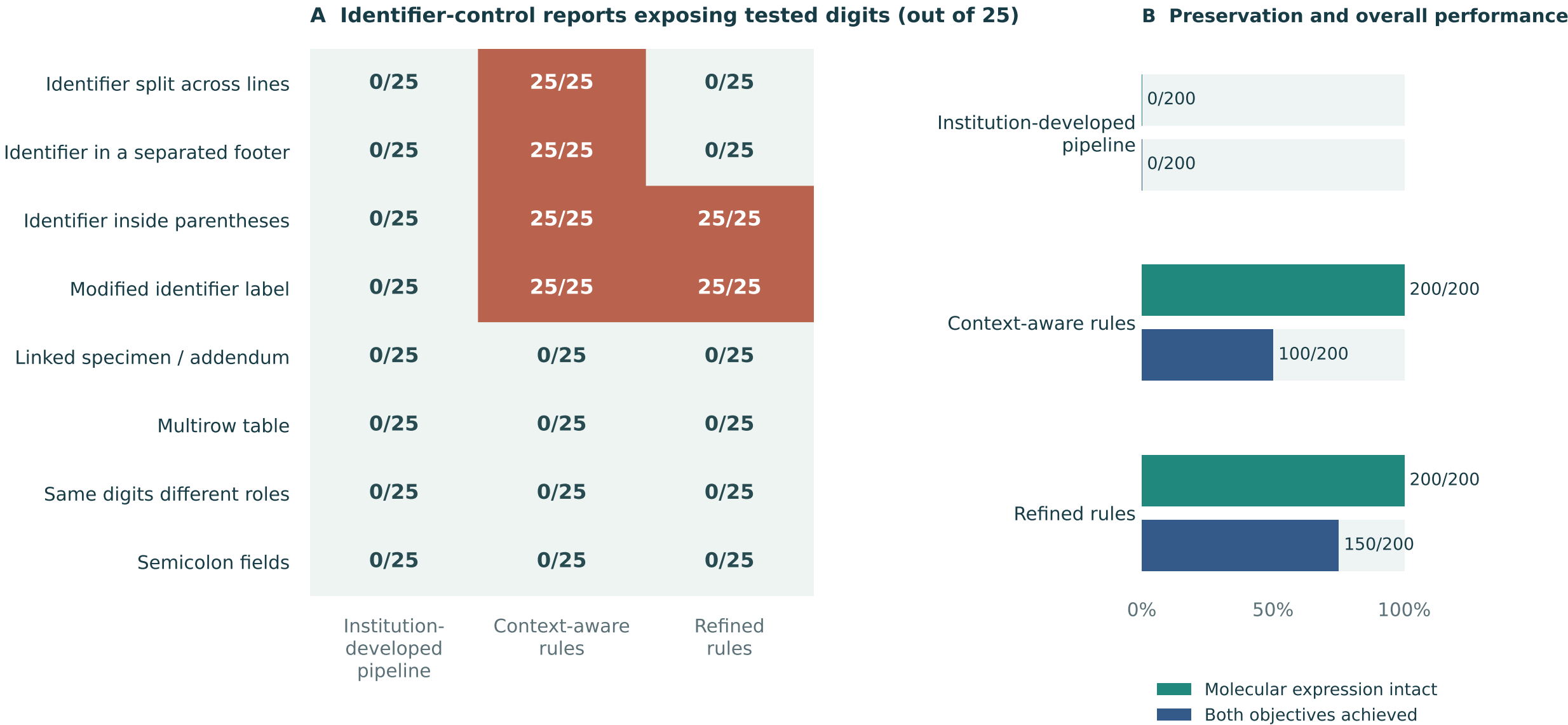
A detailed synthetic biopsy and linked EGFR report, processed by unchanged local methods.

Original synthetic report

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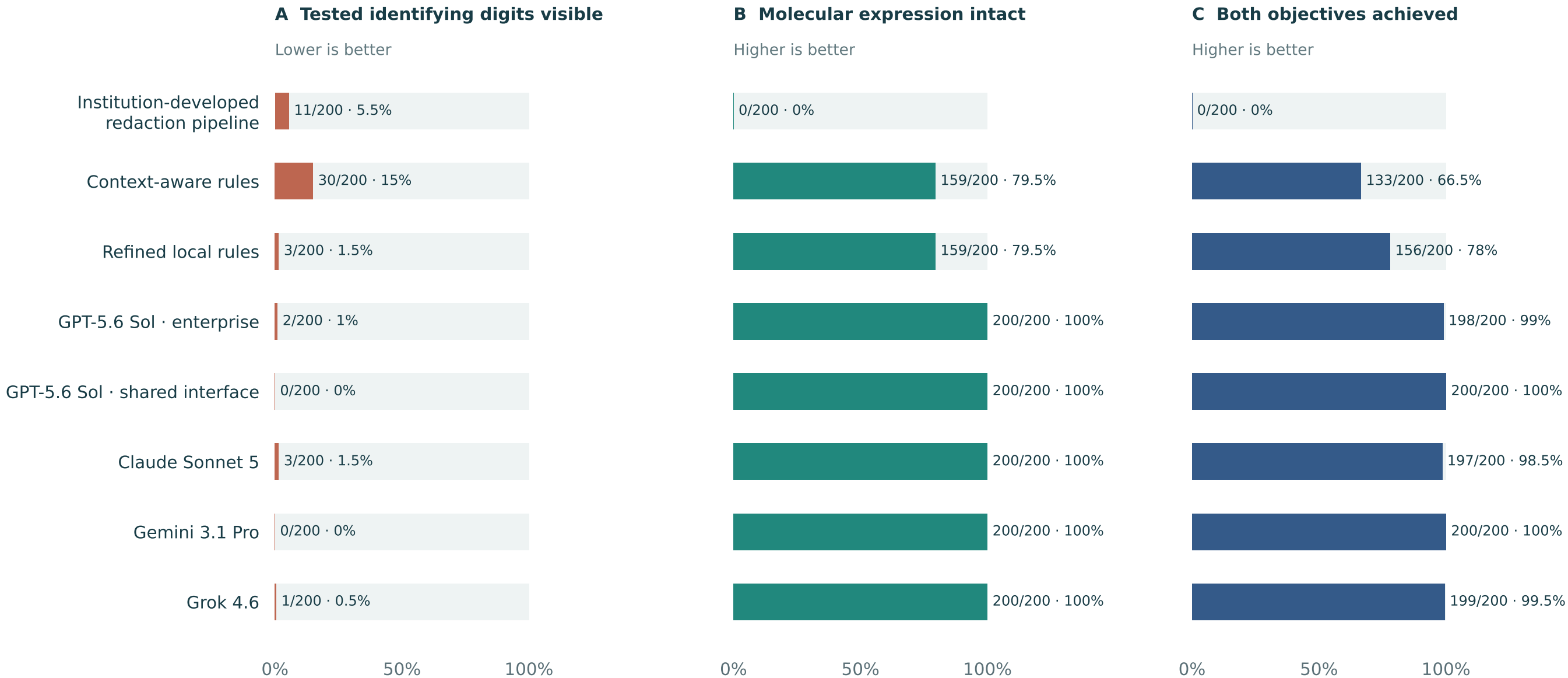
3. Local refinement: improvements and persistent failures

Separate experiment: 200 unique synthetic pairs · eight designed report patterns · 25 pairs per pattern (20 original + 5 additional)



S1. Identifier removal and molecular preservation

Main comparison: 200 unique synthetic pairs · 400 reports per approach



GPT Enterprise used the chat interface; all four other configurations used the same shared model-access interface.
Both objectives = no tested identifying-digit exposure in the control + complete molecular-expression retention in its paired report.